Form C

Core Focus on Proportions & Probability

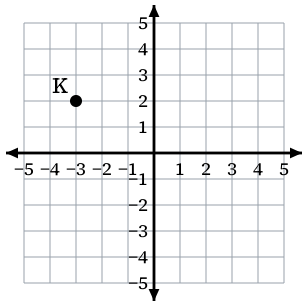
Block 5 Review ~ Direct Variation

ANSWER KEY

Name _____ Period _____ Date _____

Part I – Selected Response

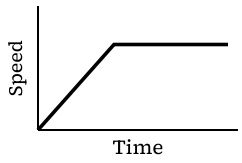
1. Which ordered pair represents the point graphed?

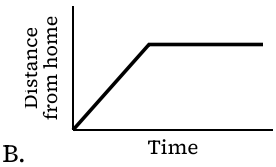


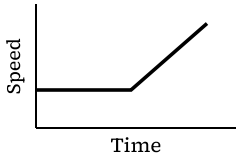
A. (3, −2)
B. (2, −3)
C. (−3, 2)
D. (−2, 3)

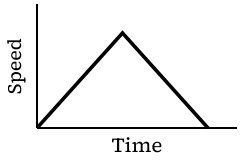
C. (−3, 2): start at the origin, move 3 left, then 2 up.

2. Which graph shows a car speeding up and then driving at a constant speed?

A.


B.


C.


D.


A. On a speed graph, rising means speeding up and a flat segment means a constant speed. (B's flat part means the car is stopped.)

3. If the values in the table below model direct variation, what is the rate?

x	0	1	2	3	4
y	0	7	14	21	28

A. $\frac{1}{7}$
B. 14
C. 7
D. It does not model direct variation.

C. 7 (y ÷ x = 7 ÷ 1 = 14 ÷ 2 = 21 ÷ 3 = 28 ÷ 4 = 7)

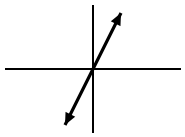
4. Which equation is the direct variation equation for the values in the table below?

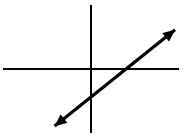
x	0	2	4	6	8
y	0	10	20	30	40

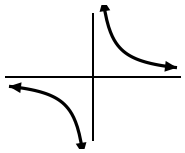
A. y = 10x
B. y = 5x
C. y = 2x
D. y = $\frac{1}{5}$x

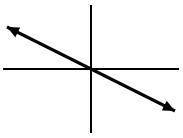
B. y = 5x (m = y ÷ x = 10 ÷ 2 = 5; check 40 ÷ 8 = 5)

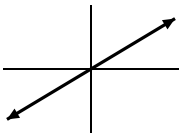
5. Which graphs show direct variation? Circle all that apply.

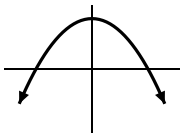
A.


B.


C.


D.


E.


F.


A, D, E. Each is a straight line through the origin. B is a line that misses the origin; C and F are curves.

6. Which equations are direct variation equations? Circle all that apply.

A. y = x − 5
B. y = 9x
C. y = $\frac{10}{x}$
D. y = $-\frac{3}{4}$x
E. y = 2^x
F. y = $\frac{x}{3}$

B, D, F. Each has the form y = mx (m = 9, −¾, ⅓). A subtracts 5, C has x in the denominator, E has x as an exponent.

7. What is the slope of the direct variation equation y = 0.6x?

A. 6

B. $\frac{3}{5}$

C. $\frac{1}{6}$

D. $\frac{5}{3}$

B. 3/5 (the slope is the number multiplying x: 0.6 = 6/10 = 3/5)

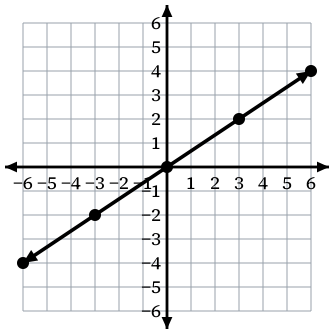
8. What is the direct variation equation for the line graphed below?

A. $y = \frac{3}{2}x$

B. $y = -\frac{2}{3}x$

C. $y = 2x$

D. $y = \frac{2}{3}x$



D. y = ⅔x (the line passes through (0, 0) and (3, 2): m = 2 ÷ 3 = ⅔)

Form C

Core Focus on Proportions & Probability

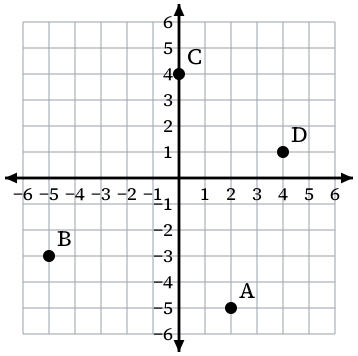
Block 5 Review ~ Direct Variation

ANSWER KEY

Name _____ Period _____ Date _____

Part II – Constructed Response

Write the ordered pair for each point on the coordinate plane to the right.



1. A

(2, −5)

2. B

(−5, −3)

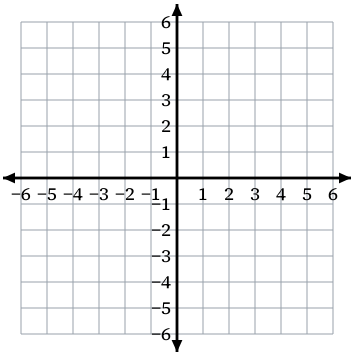
3. C

(0, 4) — on the y-axis

4. D

(4, 1)

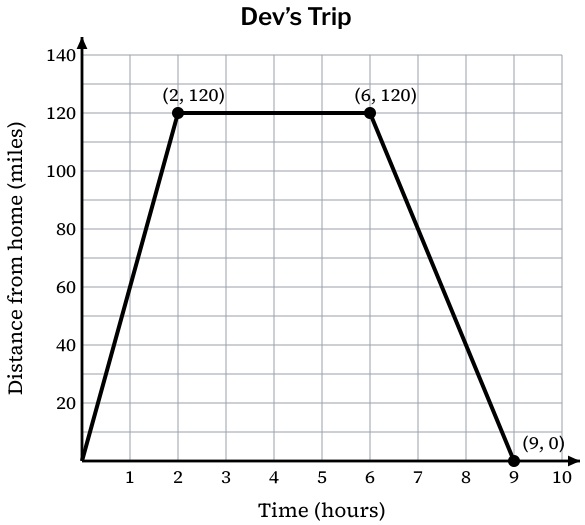
5. Plot the points (−4, 3), (4, 3) and (0, −3) and connect them in order. Connect the last point listed to the first point. What shape do the points create?



Shape: _____

Triangle. Points (−4, 3), (4, 3), (0, −3): a flat top side 8 units long and a bottom corner at (0, −3).

6. Dev's family drove to a lake, spent time there, and then drove back home. The graph below shows their distance from home (miles) over time (hours). For each part, explain how you know your answer is correct.



a. How far away is the lake from Dev's house?

120 miles. The flat part of the graph is at 120 miles.

b. How long did the family stay at the lake?

4 hours. The graph is flat from hour 2 to hour 6: 6 − 2 = 4.

c. Find the family's rate of speed in miles per hour for the drive to the lake.

60 miles per hour. 120 mi ÷ 2 h = 60.

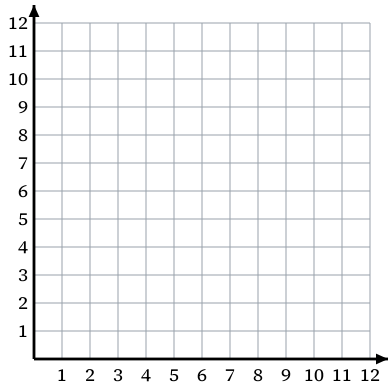
d. Find the family's rate of speed in miles per hour for the drive home.

40 miles per hour. They drive 120 mi from hour 6 to hour 9: 120 ÷ 3 = 40.

Complete each input-output table and graph each function in Quadrant I. What is the slope of each direct variation equation?

7.

Input x	Function Rule $y = \frac{1}{3}x$	Output y
0		
3		
6		
9		
12		

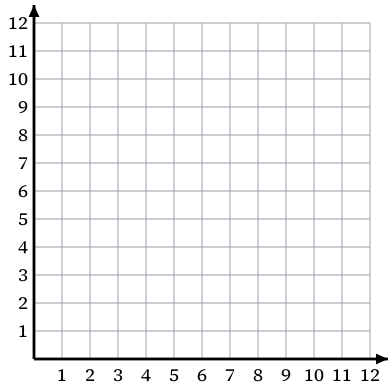


slope: _____

Outputs: 0, 1, 2, 3, 4 (⅓(0) = 0; ⅓(3) = 1; ⅓(6) = 2; ⅓(9) = 3; ⅓(12) = 4). Graph: points (0, 0), (3, 1), (6, 2), (9, 3), (12, 4) on a straight line through the origin. Slope = ⅓.

8.

Input x	Function Rule $y = \frac{3}{2}x$	Output y
0		
2		
4		
6		
8		



slope: _____

Outputs: 0, 3, 6, 9, 12 (3/2(0) = 0; 3/2(2) = 3; 3/2(4) = 6; 3/2(6) = 9; 3/2(8) = 12). Graph: points (0, 0), (2, 3), (4, 6), (6, 9), (8, 12) on a straight line through the origin. Slope = 3/2 (or 1.5).

Determine whether each table models direct variation. Circle YES or NO. If it does, give the rate. If not, explain why not.

9.

x	y
−4	−2
−2	−1
0	0
2	1
4	2

direct variation? **YES** **NO**

rate/explanation:

YES; rate ½. y ÷ x = ½ for every pair except (0, 0): −2 ÷ (−4) = ½, −1 ÷ (−2) = ½, 1 ÷ 2 = ½, 2 ÷ 4 = ½.

10.

x	y
−2	−10
−1	−5
0	0
1	5
2	12

direct variation? **YES** **NO**

rate/explanation:

NO. The ratios y ÷ x are not all equal: −10 ÷ (−2) = 5, −5 ÷ (−1) = 5, 5 ÷ 1 = 5, but 12 ÷ 2 = 6.

Determine whether each equation models direct variation. Circle YES or NO. If it does, give the rate. If not, explain why not.

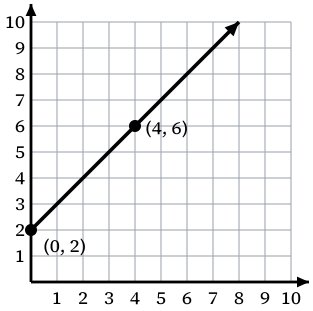
11. $y = 6x^2$ direct variation? **YES** **NO** rate/explanation:

NO. The x is squared, so the graph is a curve, not a straight line.

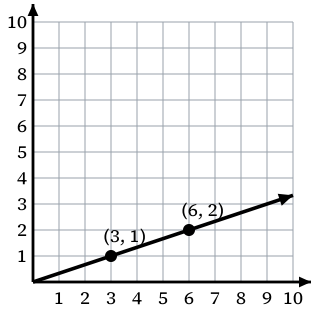
12. $y = \frac{x}{5}$ direct variation? **YES** **NO** rate/explanation:

YES; rate ⅕. y = x/5 is the same as y = ⅕x, which has the form y = mx with m = ⅕ (or 0.2).

13. Ken says Graph A models direct variation because it is a straight line and has a rate of 1. Rosa says Graph B models direct variation and has a rate of $\frac{1}{3}$. Which person is correct? Explain your reasoning.



Graph A



Graph B

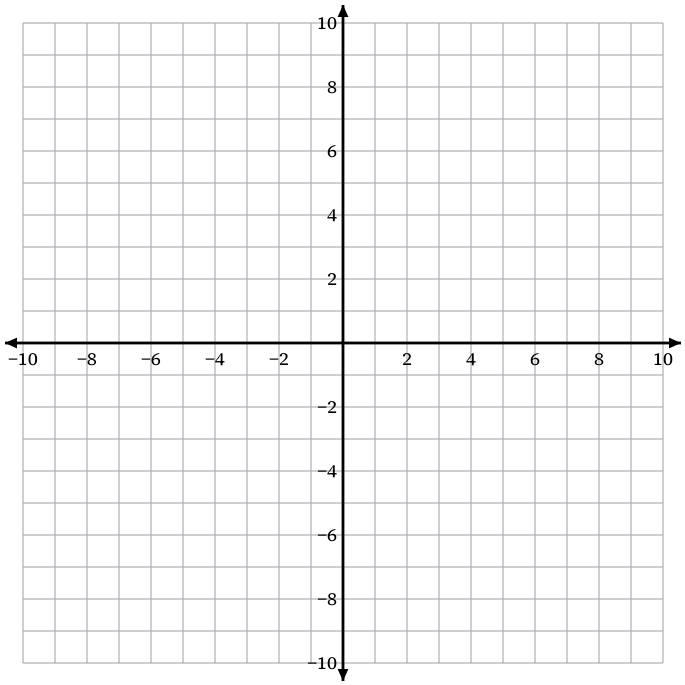
Rosa is correct. Graph B is a straight line through (0, 0), and y ÷ x = 1 ÷ 3 = 2 ÷ 6 = ⅓. Graph A is a straight line, but it starts at (0, 2), so it does not pass through the origin and is not direct variation.

14. Tickets to the school play cost $6 each. The table below shows the total cost for the number of tickets bought. Write an equation that shows the total cost, y, of x tickets.

x	y
0	0
3	18
6	36
9	54
12	72

y = 6x (m = y ÷ x = 18 ÷ 3 = 36 ÷ 6 = 6)

15. Graph the direct variation equation $y = 3x$.



A straight line through the origin that rises from left to right with slope 3: it passes through (−3, −9), (−2, −6), (−1, −3), (0, 0), (1, 3), (2, 6) and (3, 9), in Quadrants III and I.

Core Focus on Proportions & Probability

Block 5 Review ~ Direct Variation

ANSWER KEY

Name _____ Period _____ Date _____

Part III – Problem Solving

The table below shows the amount of money Marcus earned walking dogs in his neighborhood.

Hours Walking Dogs	Money Earned ($)
0	0.00
2	23.00
4	46.00
6	69.00
8	92.00

How many hours will Marcus need to walk dogs to earn $161.00?

14 hours. Rate = 23.00 ÷ 2 = $11.50 per hour (46 ÷ 4 = 11.50 too), so y = 11.5x. 161 = 11.5x → x = 161 ÷ 11.5 = 14. Check: 14 × 11.50 = 161.00.